PROGRAMMING ASSIGNMENT REPORT

Tasks 1–12

Student Name: _____

Course / Programme: _____

Institution: _____

GitHub Repository URL: _____

Important: This is a report template. Replace all prompts with your actual task work, diagrams, test results, and complexity analysis before uploading.

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Include page numbers for the final report sections after completing the report.

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Task 1: [Insert Task Title]

1. Problem Statement

Paste the exact task question and briefly state the objective.

2. Approach / Algorithm

Explain the solution approach and key decisions in your own words.

3. Pseudocode

Add step-by-step pseudocode for this task.

4. UML Diagram (if required/applicable)

Insert the relevant UML diagram and explain its main elements.

5. Test Table

Test ID	Input	Expected Output	Actual Output	Result
TC-01	TODO	TODO	TODO	Pass/Fail
TC-02	TODO	TODO	TODO	Pass/Fail

6. Complexity Analysis

Time complexity: TODO. Explain why. Space complexity: TODO. Explain why.

7. Execution Evidence

Insert required output screenshot(s) and describe the result. Ensure screenshots match the submitted code.

8. Summary

State whether the task works as expected and note any limitations.

Task 2: [Insert Task Title]

1. Problem Statement

Paste the exact task question and briefly state the objective.

2. Approach / Algorithm

Explain the solution approach and key decisions in your own words.

3. Pseudocode

Add step-by-step pseudocode for this task.

4. UML Diagram (if required/applicable)

Insert the relevant UML diagram and explain its main elements.

5. Test Table

Test ID	Input	Expected Output	Actual Output	Result
TC-01	TODO	TODO	TODO	Pass/Fail
TC-02	TODO	TODO	TODO	Pass/Fail

6. Complexity Analysis

Time complexity: TODO. Explain why. Space complexity: TODO. Explain why.

7. Execution Evidence

Insert required output screenshot(s) and describe the result. Ensure screenshots match the submitted code.

8. Summary

State whether the task works as expected and note any limitations.

Task 3: [Insert Task Title]

1. Problem Statement

Paste the exact task question and briefly state the objective.

2. Approach / Algorithm

Explain the solution approach and key decisions in your own words.

3. Pseudocode

Add step-by-step pseudocode for this task.

4. UML Diagram (if required/applicable)

Insert the relevant UML diagram and explain its main elements.

5. Test Table

Test ID	Input	Expected Output	Actual Output	Result
TC-01	TODO	TODO	TODO	Pass/Fail
TC-02	TODO	TODO	TODO	Pass/Fail

6. Complexity Analysis

Time complexity: TODO. Explain why. Space complexity: TODO. Explain why.

7. Execution Evidence

Insert required output screenshot(s) and describe the result. Ensure screenshots match the submitted code.

8. Summary

State whether the task works as expected and note any limitations.

Task 4: [Insert Task Title]

1. Problem Statement

Paste the exact task question and briefly state the objective.

2. Approach / Algorithm

Explain the solution approach and key decisions in your own words.

3. Pseudocode

Add step-by-step pseudocode for this task.

4. UML Diagram (if required/applicable)

Insert the relevant UML diagram and explain its main elements.

5. Test Table

Test ID	Input	Expected Output	Actual Output	Result
TC-01	TODO	TODO	TODO	Pass/Fail
TC-02	TODO	TODO	TODO	Pass/Fail

6. Complexity Analysis

Time complexity: TODO. Explain why. Space complexity: TODO. Explain why.

7. Execution Evidence

Insert required output screenshot(s) and describe the result. Ensure screenshots match the submitted code.

8. Summary

State whether the task works as expected and note any limitations.

Task 5: [Insert Task Title]

1. Problem Statement

Paste the exact task question and briefly state the objective.

2. Approach / Algorithm

Explain the solution approach and key decisions in your own words.

3. Pseudocode

Add step-by-step pseudocode for this task.

4. UML Diagram (if required/applicable)

Insert the relevant UML diagram and explain its main elements.

5. Test Table

Test ID	Input	Expected Output	Actual Output	Result
TC-01	TODO	TODO	TODO	Pass/Fail
TC-02	TODO	TODO	TODO	Pass/Fail

6. Complexity Analysis

Time complexity: TODO. Explain why. Space complexity: TODO. Explain why.

7. Execution Evidence

Insert required output screenshot(s) and describe the result. Ensure screenshots match the submitted code.

8. Summary

State whether the task works as expected and note any limitations.

Task 6: [Insert Task Title]

1. Problem Statement

Paste the exact task question and briefly state the objective.

2. Approach / Algorithm

Explain the solution approach and key decisions in your own words.

3. Pseudocode

Add step-by-step pseudocode for this task.

4. UML Diagram (if required/applicable)

Insert the relevant UML diagram and explain its main elements.

5. Test Table

Test ID	Input	Expected Output	Actual Output	Result
TC-01	TODO	TODO	TODO	Pass/Fail
TC-02	TODO	TODO	TODO	Pass/Fail

6. Complexity Analysis

Time complexity: TODO. Explain why. Space complexity: TODO. Explain why.

7. Execution Evidence

Insert required output screenshot(s) and describe the result. Ensure screenshots match the submitted code.

8. Summary

State whether the task works as expected and note any limitations.

Task 7: [Insert Task Title]

1. Problem Statement

Paste the exact task question and briefly state the objective.

2. Approach / Algorithm

Explain the solution approach and key decisions in your own words.

3. Pseudocode

Add step-by-step pseudocode for this task.

4. UML Diagram (if required/applicable)

Insert the relevant UML diagram and explain its main elements.

5. Test Table

Test ID	Input	Expected Output	Actual Output	Result
TC-01	TODO	TODO	TODO	Pass/Fail
TC-02	TODO	TODO	TODO	Pass/Fail

6. Complexity Analysis

Time complexity: TODO. Explain why. Space complexity: TODO. Explain why.

7. Execution Evidence

Insert required output screenshot(s) and describe the result. Ensure screenshots match the submitted code.

8. Summary

State whether the task works as expected and note any limitations.

Task 8: [Insert Task Title]

1. Problem Statement

Paste the exact task question and briefly state the objective.

2. Approach / Algorithm

Explain the solution approach and key decisions in your own words.

3. Pseudocode

Add step-by-step pseudocode for this task.

4. UML Diagram (if required/applicable)

Insert the relevant UML diagram and explain its main elements.

5. Test Table

Test ID	Input	Expected Output	Actual Output	Result
TC-01	TODO	TODO	TODO	Pass/Fail
TC-02	TODO	TODO	TODO	Pass/Fail

6. Complexity Analysis

Time complexity: TODO. Explain why. Space complexity: TODO. Explain why.

7. Execution Evidence

Insert required output screenshot(s) and describe the result. Ensure screenshots match the submitted code.

8. Summary

State whether the task works as expected and note any limitations.

Task 9: [Insert Task Title]

1. Problem Statement

Paste the exact task question and briefly state the objective.

2. Approach / Algorithm

Explain the solution approach and key decisions in your own words.

3. Pseudocode

Add step-by-step pseudocode for this task.

4. UML Diagram (if required/applicable)

Insert the relevant UML diagram and explain its main elements.

5. Test Table

Test ID	Input	Expected Output	Actual Output	Result
TC-01	TODO	TODO	TODO	Pass/Fail
TC-02	TODO	TODO	TODO	Pass/Fail

6. Complexity Analysis

Time complexity: TODO. Explain why. Space complexity: TODO. Explain why.

7. Execution Evidence

Insert required output screenshot(s) and describe the result. Ensure screenshots match the submitted code.

8. Summary

State whether the task works as expected and note any limitations.

Task 10: [Insert Task Title]

1. Problem Statement

Paste the exact task question and briefly state the objective.

2. Approach / Algorithm

Explain the solution approach and key decisions in your own words.

3. Pseudocode

Add step-by-step pseudocode for this task.

4. UML Diagram (if required/applicable)

Insert the relevant UML diagram and explain its main elements.

5. Test Table

Test ID	Input	Expected Output	Actual Output	Result
TC-01	TODO	TODO	TODO	Pass/Fail
TC-02	TODO	TODO	TODO	Pass/Fail

6. Complexity Analysis

Time complexity: TODO. Explain why. Space complexity: TODO. Explain why.

7. Execution Evidence

Insert required output screenshot(s) and describe the result. Ensure screenshots match the submitted code.

8. Summary

State whether the task works as expected and note any limitations.

Task 11: [Insert Task Title]

1. Problem Statement

Paste the exact task question and briefly state the objective.

2. Approach / Algorithm

Explain the solution approach and key decisions in your own words.

3. Pseudocode

Add step-by-step pseudocode for this task.

4. UML Diagram (if required/applicable)

Insert the relevant UML diagram and explain its main elements.

5. Test Table

Test ID	Input	Expected Output	Actual Output	Result
TC-01	TODO	TODO	TODO	Pass/Fail
TC-02	TODO	TODO	TODO	Pass/Fail

6. Complexity Analysis

Time complexity: TODO. Explain why. Space complexity: TODO. Explain why.

7. Execution Evidence

Insert required output screenshot(s) and describe the result. Ensure screenshots match the submitted code.

8. Summary

State whether the task works as expected and note any limitations.

Task 12: [Insert Task Title]

1. Problem Statement

Paste the exact task question and briefly state the objective.

2. Approach / Algorithm

Explain the solution approach and key decisions in your own words.

3. Pseudocode

Add step-by-step pseudocode for this task.

4. UML Diagram (if required/applicable)

Insert the relevant UML diagram and explain its main elements.

5. Test Table

Test ID	Input	Expected Output	Actual Output	Result
TC-01	TODO	TODO	TODO	Pass/Fail
TC-02	TODO	TODO	TODO	Pass/Fail

6. Complexity Analysis

Time complexity: TODO. Explain why. Space complexity: TODO. Explain why.

7. Execution Evidence

Insert required output screenshot(s) and describe the result. Ensure screenshots match the submitted code.

8. Summary

State whether the task works as expected and note any limitations.

Conclusion

Summarise what was implemented across Tasks 1–12. Replace this prompt with your own conclusion.